RESEARCH PAPER

Ferry Capacity Utilization & Operational Efficiency Analytics System

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Abstract

This paper analyzes ferry capacity utilization and operational efficiency for the Toronto Island Ferry system using dashboard insights from 2015 to 2025. Key performance indicators (KPIs) such as average utilization, load factor, waste ratio, and stability are examined to understand seasonal demand, congestion patterns, and long-term operational trends. The findings highlight areas of efficiency and opportunities for optimization in public transport management.

Introduction

Urban ferry systems play a critical role in connecting communities and supporting tourism. Efficient management of ferry capacity ensures passenger satisfaction, reduces congestion, and minimizes operational waste. This study uses data visualizations and KPIs from the Toronto Island Ferry analytics dashboard to evaluate system performance over a ten-year period.

Methodology

Data was collected and visualized through a Streamlit-based dashboard. The analysis window spans January 2015 to December 2025, with a time resolution of 15 minutes. Seasonal filters (Off Season and Summer) and a congestion threshold of 75% were applied to identify utilization patterns.

Dashboard Visualization

The figure below shows the main dashboard of the Ferry Intelligence Platform, displaying all key metrics and charts used in this analysis:

Key Performance Indicators

- **Average Utilization:** 0.9% Indicates very low overall usage relative to capacity, suggesting significant spare capacity.
- **Average Load Factor:** 1.20 Shows that ferries often carry more passengers than expected per trip, pointing to uneven distribution of demand.
- **Waste Ratio (Capacity):** 3.4% Reflects minimal wasted capacity, implying efficient scheduling despite low utilization.
- **Total Loads (Volume):** 23,932,470 Demonstrates the scale of passenger movement over the study period.
- **Stability (Consistency):** 98.4 High consistency in operations, showing reliable service delivery.

Results

1. **Utilization Trends (2015–2025):** Peaks observed around 2018 and 2024, likely linked to tourism surges or special events. Otherwise, utilization remains low.
2. **Seasonal Distribution:** Summer shows higher variability in utilization compared to Off Season, confirming seasonal demand fluctuations.
3. **Weekly Congestion Heatmap:** Midday hours exhibit higher congestion, while mornings and evenings remain underutilized. This suggests opportunities for demand redistribution.

Discussion

The data reveals a paradox: while utilization is low overall, congestion occurs at specific times and seasons. This imbalance highlights the need for dynamic scheduling, targeted marketing during off-peak hours, and possibly adjusting ferry sizes to better match demand. The high stability score indicates strong operational reliability, which can be leveraged to experiment with flexible service models.

Conclusion

The Toronto Island Ferry system demonstrates reliable operations but faces challenges in balancing capacity and demand. By addressing seasonal peaks and midday congestion, the system can improve efficiency and passenger experience. Future research should explore predictive modeling and integration with tourism data to further optimize ferry utilization.

References & Data Sources

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